

Chapter -5

Market Basket Analysis (MBA)

- Market Basket Analysis is a data mining technique used to find relationships between items purchased together.
Example: If customers buy **bread**, they are likely to buy **butter** too.
- It helps retailers understand buying habits and make decisions like product placement, cross-selling, and promotions.

- **Frequent Itemsets**

- - A *frequent itemset* is a set of items that appear together frequently in transactions.
Example: {Milk, Bread} might appear in 40% of transactions.

- **Support:**

The *support* of an itemset is the proportion (or number) of transactions that contain it.

$$\text{Support}(A) = \frac{\text{Number of transactions containing } A}{\text{Total transactions}}$$

- **Closed Itemsets**

- **Definition:**

A *closed itemset* is a frequent itemset that has no *superset* with the same support count.

In other words, it's a **maximal** version of a frequent itemset with identical support.

- **Example:**

Suppose {Milk, Bread} and {Milk, Bread, Butter} have the same support.

Then {Milk, Bread} is **not closed** (since {Milk, Bread, Butter} has the same support),

but {Milk, Bread, Butter} **is closed**.

- **Association Rules**

- **Definition:**

Association rules show the relationship between items in transactions, written as:

- $A \Rightarrow B$ Meaning: “If A is bought, B is likely to be bought.”

- **Metrics:**

- **Support:** How often A and B appear together.
- **Confidence:** Probability that B is bought given A is bought.

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$$\text{Confidence}(A \Rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

- **Lift:** Strength of association.

$$\text{Lift}(A \Rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A) \times \text{Support}(B)}$$

- If **Lift** > 1, A and B are positively related.
- If **Lift** = 1, A and B are independent.

- **Frequent Pattern Mining**

- **Definition:**

The process of finding patterns or itemsets that occur frequently in a dataset.

Market Basket Analysis is one application of frequent pattern mining.

- **Goal:**

To discover hidden patterns and associations from large transactional data.

Apriori Algorithm

- The **Apriori Algorithm** is one of the most popular algorithms in **data mining** for **Association Rule Learning**.
It is used to find **frequent itemsets** (sets of items that occur frequently together in transactions) and then derive **association rules** from them.
- It is called “**Apriori**” because it uses **prior knowledge** of frequent itemset properties —
that is, **any subset of a frequent itemset must also be frequent** (called the **Apriori Property**).

- **Imagine a small shop**
- People are buying some grocery items.
Here are the **5 transactions** recorded:

Transaction	Items Bought
T1	Milk, Bread
T2	Milk, Bread, Butter
T3	Milk, Butter
T4	Bread, Butter
T5	Milk, Bread, Butter

- We want to find which items are **bought together** most often. This will help the shopkeeper know —
- “If someone buys *Milk*, are they also likely to buy *Bread* or *Butter*?”
- **Step 1: Count each item (1-itemsets)**
- We count how many times each item appears in all transactions.

Item	Times Bought
Milk	4
Bread	4
Butter	4

All appear in 4 out of 5 transactions → all are **frequent** items.

- **Step 2: Make pairs (2-itemsets)**
- Now, we check which **pairs of items** are often bought together.

Pair	How many times they appear together
Milk & Bread	3 (T1, T2, T5)
Milk & Butter	3 (T2, T3, T5)
Bread & Butter	3 (T2, T4, T5)

Each pair appears 3 times → all are **frequent pairs**.

- **tep 3: Try triplets (3-itemsets)**
- Now, we check if all three items are bought together.

Triplet

Milk, Bread, Butter

How many times

2 (T2, T5)

Only 2 out of 5 → not frequent enough (below 60%).
So we stop here.

Frequent Items

{Milk}

{Bread}

{Butter}

{Milk, Bread}

{Milk, Butter}

{Bread, Butter}

How often

4

4

4

3

3

3

- **Step 4: Make Association Rules**
- We now find “**if-then**” relationships.
- **Example Rule 1:**
- **If a customer buys Milk → they also buy Bread**
- Support = 3 out of 5 = **60%**
- Confidence = 3 (both) ÷ 4 (milk buyers) = **75%**
- Meaning:
- 75% of people who bought *Milk* also bought *Bread*.

- **Example Rule 2:**
- **If a customer buys Bread → they also buy Butter**
- Support = $3/5 = 60\%$
- Confidence = $3 \div 4 = 75\%$
- Meaning:
- 75% of people who bought *Bread* also bought *Butter*.

- People often buy **Milk + Bread + Butter** together.
- The most common pairs are:
- Milk & Bread
- Bread & Butter
- Milk & Butter

Apriori algorithm helps find:

- “Which products are bought together often.”

- **Real-Life Use**

- Amazon: “People who bought this also bought that.”
- Supermarkets: Product placement (keep Milk and Bread near each other).
- Online shopping: Recommendation systems.

- **What is Association Rule Generation?**
- Association Rule Generation is the **final step** of the **Apriori Algorithm**. Once we find which items are **frequently bought together** (**frequent itemsets**), we then create **rules** that show **how items are related**.
- It means finding patterns like:
“If a person buys **Milk**, they are also likely to buy **Bread**.”
- This is written as a rule:
- $\text{Milk} \Rightarrow \text{Bread}$. (“If Milk, then Bread”)

Example

Transaction

T1

T2

T3

T4

T5

Items Bought

Milk, Bread

Milk, Bread, Butter

Milk, Butter

Bread, Butter

Milk, Bread, Butter

Step 1: Find Frequent Itemsets

From Apriori, we already know:

- {Milk, Bread} → appears 3 times
- {Milk, Butter} → appears 3 times
- {Bread, Butter} → appears 3 times

Total transactions = 5

- **Step 2: Create Possible Rules**

- From {Milk, Bread}, we can make two rules:

1. Milk $\rightarrow$ Bread

2. Bread $\rightarrow$ Milk

- **Step 3: Calculate Metrics**

- **Rule 1: Milk $\rightarrow$ Bread**

- Support = $3/5 = 60\%$

- Confidence = $3/4 = 75\%$

75% of people who bought *Milk* also bought *Bread*.

- **Rule 2: Bread → Milk**
- Support = $3/5 = 60\%$
- Confidence = $3/4 = 75\%$

75% of people who bought *Bread* also bought *Milk*.

- **Step 4: Check Against Thresholds**
- If we decide:
- Minimum Support = 50%
- Minimum Confidence = 70%
- Then both rules qualify ✓
So they are **strong association rules**.
- ✓ **Step 5: Calculate Lift (optional but useful)**
- For rule **Milk → Bread**:

$$\begin{aligned}\text{Lift} &= \frac{\text{Support}(\text{Milk}, \text{Bread})}{\text{Support}(\text{Milk}) \times \text{Support}(\text{Bread})} \\ &= \frac{0.6}{0.8 \times 0.8} = 0.9375\end{aligned}$$

ift < 1 → Milk and Bread are slightly **negatively related** (buying Milk doesn't strongly increase chance of Bread).

- ◆ If **Lift > 1**, it means **strong positive relationship**.

Rule	Meaning	Interpretation
Milk → Bread	If someone buys Milk, they also buy Bread	75% confidence means this happens often
Bread → Butter	If someone buys Bread, they also buy Butter	Also likely association
Milk → Butter	If someone buys Milk, they also buy Butter	75% confidence too

- **Why Association Rules Are Useful**
 - **Retail:** Product placement & combo offers
 - **E-commerce:** “Frequently bought together” suggestions
 - **Restaurants:** Find meal combinations
- Banking:** Detect common spending patterns

Step

1

2

3

4

5

Description

Find frequent itemsets using Apriori

Make possible “If-Then” rules

Calculate Support & Confidence

Keep rules that meet the thresholds

(Optional) Measure Lift to find strongest rules

Rule	Support	Confidence	Lift	Result
Milk → Bread	60%	75%	0.93	Strong
Bread → Butter	60%	75%	0.93	Strong
Milk → Butter	60%	75%	0.93	Strong